A Pure Analytical Gaussian Expansion Method for Heavy and Mixed-State Quarkonia

Mass spectra and decay widths of $b\bar{b}$, $c\bar{c}$, and B_c

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Outline

1. Motivation & Theoretical Framework

Motivation: Quarkonia as a QCD Laboratory

- **Objective:** Heavy quarkonia — bound states of a heavy quark and its antiquark ($c\bar{c}$, $b\bar{b}$) — a clean, non-relativistic probe of the **confining, low-energy regime** of QCD where perturbation theory fails.
- **Why non-relativistic:** the heavy quarks are *slow* ($v^2/c^2 \approx 0.3$ for charm, 0.1 for bottom), so a Schrödinger picture with a static potential is well justified.
- **Scope:** a *single* framework spanning **three sectors** — bottomonium $b\bar{b}$, charmonium $c\bar{c}$, and the mixed-flavour B_c ($b\bar{c}$) — fixed only by the two quark masses and the potential.

The Cornell Potential & Hamiltonian

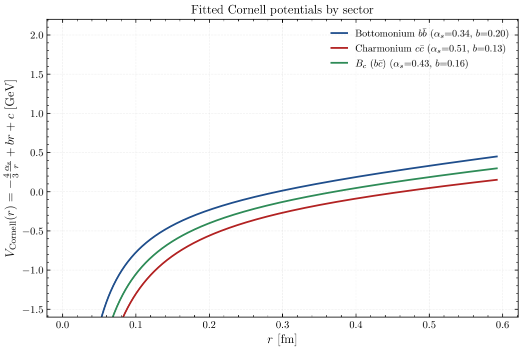
- **Model:** the phenomenological **Cornell potential** — a single-gluon Coulomb attraction plus a *linear* confining term from the QCD string:

$$V(r) = -\frac{4}{3}\frac{\alpha_s}{r} + b\,r + c$$

- **Full Hamiltonian** adds the spin-dependent Breit–Fermi fine structure:

$$H = \frac{p^2}{2\mu} + V(r) + \underbrace{V_{LS} + V_{SS} + V_T}_{\text{spin terms}}$$

- Short range $\sim$ asymptotic freedom; the linear slope $b \approx 0.18\ \mathrm{GeV}^2$ is the **string tension** — confinement.



The three fitted Cornell potentials.

2. The Gaussian Expansion Method

Methodology: The Gaussian Expansion Method

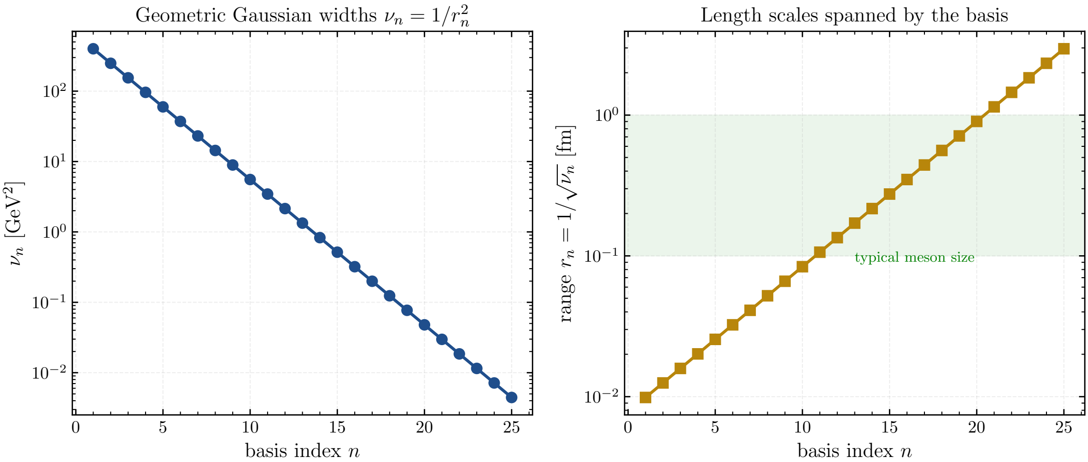
- **Core technique:** the Rayleigh–Ritz variational method in a basis of **25 Gaussians** $\phi_n(r) = r^{\,l}\, e^{-\nu_n r^2}$, with widths ν_n in *geometric progression* over $r \in [0.05, 15]\ \mathrm{GeV}^{-1}$.
- **Generalized eigenproblem:** expanding $\psi = \sum_n c_n \phi_n$ turns $H\psi = E\psi$ into

$$H\,\boldsymbol{c} = E\,S\,\boldsymbol{c},$$

  ($S =$ overlap), solved with `scipy.linalg.eigh`.
- **Why Gaussians:** one geometric mesh resolves both the short-distance Coulomb cusp and the long confined tail in a single, well-conditioned basis.

The Gaussian Width Mesh

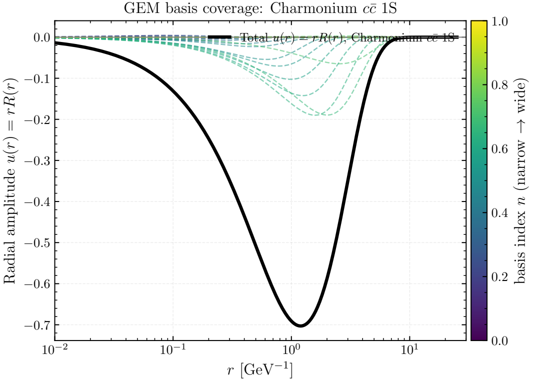
GEM geometric mesh ($n_{\max} = 25$, $r_{\min} = 0.05$, $r_{\max} = 15\ \mathrm{GeV}^{-1}$)



Geometric width mesh $\nu_n = 1/r_n^2$ and the length scales it spans.

The Analytic Engine: Closed-Form Matrix Elements

- **Key trick:** *every* element of H and S — kinetic, Coulomb, linear, and the Gaussian-smeared hyperfine term — reduces to a closed-form **Γ-function integral**. *No numerical quadrature, no radial grid.*
- **Consequence:** the *only* approximation is the finite basis size. There is no quadrature error to control — the build is fast and **exactly reproducible**.
- This is what "*pure analytical*" means: the GEM here is an algebraic machine, not a numerical integrator.



25 weighted Gaussians (dashed) summing to the charmonium $1S$ amplitude $u(r) = rR(r)$.

Wavefunctions & Convergence

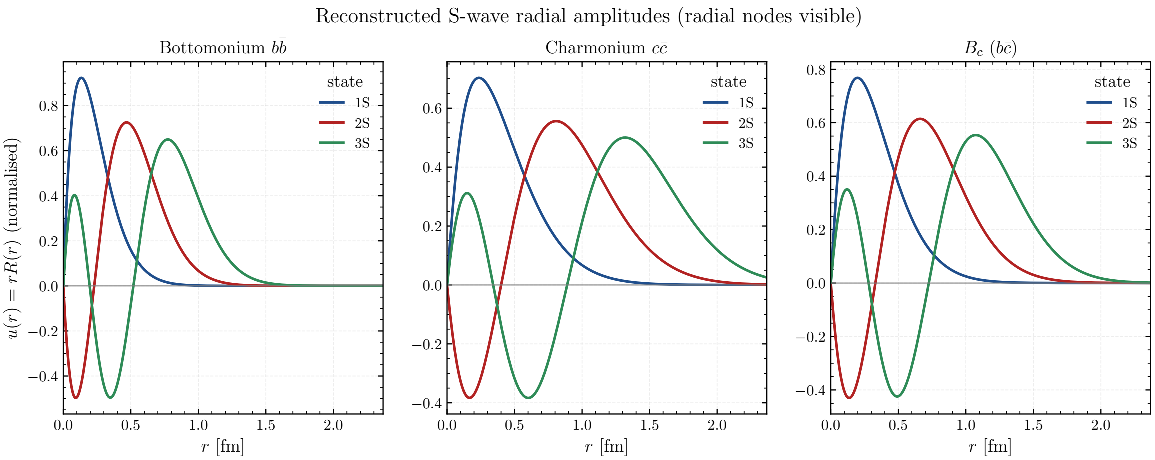
- **Output:** reconstructed radial amplitudes $u(r)$ show the correct $0/1/2$ **radial nodes** for $1S/2S/3S$ and the tighter binding of the heavier systems.
- **Convergence guarantee (virial theorem):** for a converged basis

$$\frac{\langle 2T \rangle}{\langle r\, dV/dr \rangle} = 1.00,$$

including the hyperfine term for S-waves — a *parameter-free* check that the eigenstates are physical.

- If this ratio drifts from 1, the basis is not converged and every downstream number is suspect.

Radial Wavefunctions & Nodes



S-wave amplitudes $u(r) = rR(r)$ for $1S/2S/3S$ per sector.

3. Spin-Dependent Corrections

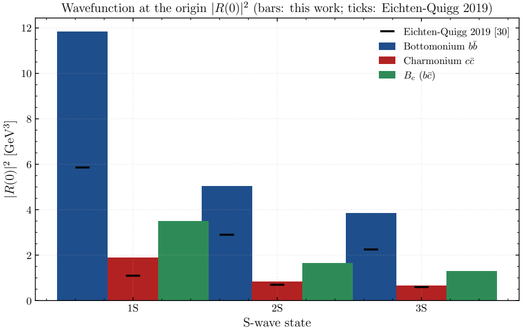
Fine Structure: Breit–Fermi Corrections

Approach: Breit–Fermi spin interactions are treated directly within the **variational method**. The spin-dependent matrix elements are evaluated exactly using the same `analytical_integral` machinery and incorporated into the full Hamiltonian.

- **Spin–orbit** V_{LS} $(L > 0)$: splits the ^3P_J and ^3D_J multiplets.
- **Spin–spin** V_{SS} (hyperfine): the contact term separating singlets from triplets — e.g. the J/ψ–η_c splitting. It is smeared by a Gaussian of width $\sigma \sim \sqrt{\mu}$ tied to the mass scale.
- **Tensor** V_T**:** the diagonal tensor splitting *within* the spin-triplet multiplets.

S–D Mixing & the Wavefunction at the Origin

- **Off-diagonal tensor (S–D mixing):** the element $\langle 2^3S_1 \,|\, V_T \,|\, 1^3D_1 \rangle$ mixes S- and D-waves — indispensable for the $\psi(3770)$, a $2S$–$1D$ admixture.
- $|R(0)|^2$ **via the hypervirial theorem:** the S-wave wavefunction-at-the-origin — which *every* annihilation width needs — is obtained as an **expectation value**, never by evaluating ψ at $r = 0$ directly. Far more numerically stable.
- These two ingredients connect the spectrum to the *decay widths* (next sections).



$|R(0)|^2$ for the S-waves — it drives all annihilation widths.

4. Fitting & Uncertainties

Parameter Fitting: A True χ^2 Objective

- **Data:** experimental masses *and* the ground / first-excited leptonic e^+e^- widths, read live from the official **PDG 2026** SQLite tables (no hand-edited inputs).
- **Objective:** a genuinely dimensionless χ^2 — each residual is $(\text{model} - \text{exp})/\sigma_{\text{physical}}$, *not* a hand-tuned weight. Folding the leptonic widths in lets the wavefunction at the origin help pin the potential — masses alone are not enough.
- **Optimizer:** `scipy.optimize.least_squares` with the trust-region-reflective method, which handles the parameter bounds and the under-determined B_c sector that plain Levenberg–Marquardt cannot.

Two Distinct Uncertainties

- **Model uncertainty (the $\pm$ on a value):** finite-difference propagation of the fitted Cornell covariance — "how far does a prediction move when the parameters wiggle within their errors."
- **Physical error bar (the σ in a χ^2 or pull):**

$$\sigma = \sqrt{\sigma^2_{\text{exp}} + \sigma^2_{\text{theory},i}}, \qquad \sigma_{\text{theory},i} = \left| \left\langle \tfrac{p^4}{8c^2}\left(m_1^{-3} + m_2^{-3}\right)\right\rangle_i \right|.$$

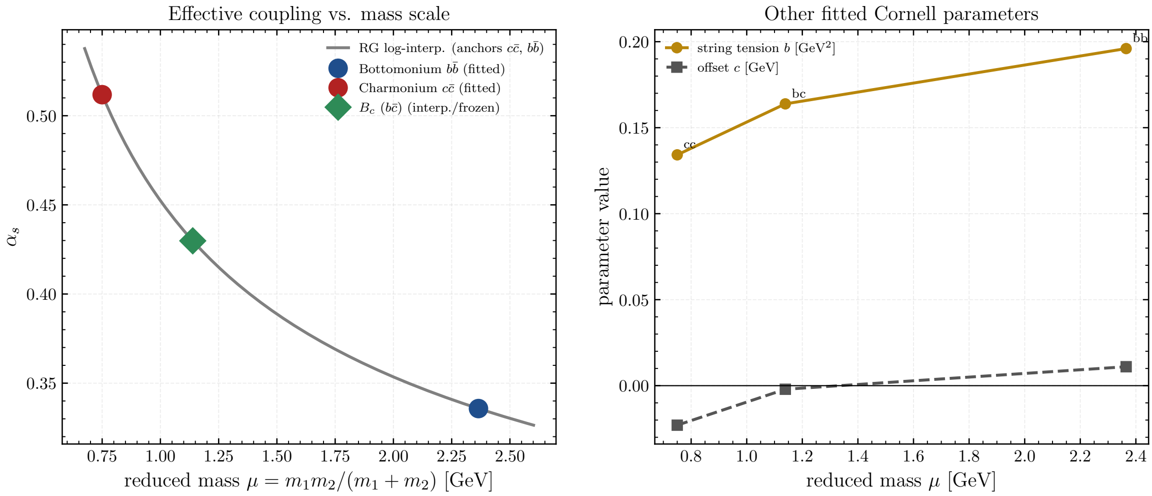
  Derived, not tuned: the leading omitted relativistic correction, computed **per state** — automatically larger for charm ($\langle v^2\rangle \approx 0.23$) than bottom (0.08). Widths get a derived per-sector fraction $\sqrt{(16\alpha_s/3\pi)^2 + \langle v^2\rangle}$ ($\approx 0.44/0.77$).

- **Why it matters:** PDG mass errors are only 0.1–1 MeV; no NR potential model can reach that. With this σ, $\chi^2/\mathrm{dof} \lesssim 1$ means "reproduces a level to within the size of the leading physics it omits." The headline figure is the assumption-free **RMS mass deviation**.

Three Sectors & the B_c Interpolation

- $b\bar{b}$ **and** $c\bar{c}$**:** fully fitted — four free Cornell parameters (α_s, b, c, σ) each.
- B_c **is special:** too few known states to pin α_s and σ. They are **log-interpolated** (running-coupling style) between the $c\bar{c}$ and $b\bar{b}$ fits at the B_c reduced-mass scale, with uncertainties propagated through the interpolation — only b and c are tuned to the two measured B_c masses.

The B_c Coupling by Interpolation



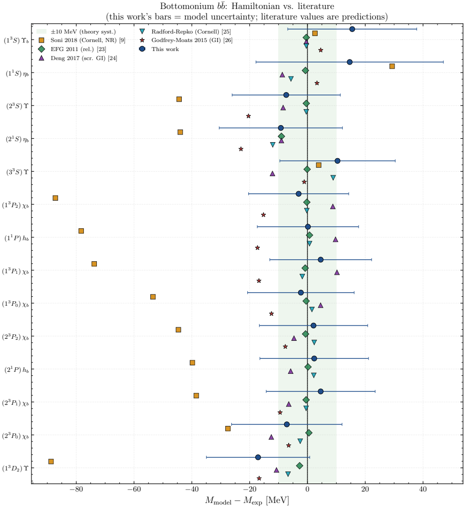
Effective α_s vs reduced-mass scale: $c\bar{c}$ and $b\bar{b}$ anchor the B_c coupling.

5. Results: Mass Spectra

Results: Bottomonium Spectrum

- **RMS deviation 8.9 MeV** — the fitted S/P levels sit *within* their derived relativistic bound (fit $\chi^2/\mathrm{dof} = 0.37$, all pulls $\lesssim 1$).
- 14 measured levels reproduced; the model then **predicts** the full D-wave tower and the $3S/3P$ levels.
- Largest single tension is the predicted 1^3D_2 ($\Upsilon_2(1D)$) at $-17\,\mathrm{MeV}$ — beyond its small ($1.6\,\mathrm{MeV}$) kinetic relativistic bar, flagging the D-wave fine structure.

Bottomonium vs. the Literature

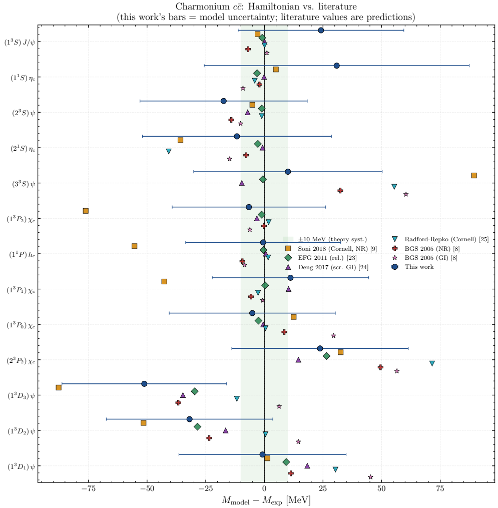


Per-state deviation $M_{\text{model}} - M_{\text{exp}}$: this work vs. five literature models; green band $= \pm 10\,\text{MeV}$ visual guide.

Results: Charmonium Spectrum

- **RMS deviation 30.2 MeV** — the fitted levels still lie within their bound (fit $\chi^2/\mathrm{dof} = 0.22$), but the predicted high-L states exceed it, and *that is physics*.
- Charm is lighter, $\langle v^2 \rangle \approx 0.23$, so the neglected $\mathcal{O}(v^2/c^2)$ corrections *exceed* the leading kinetic estimate — concentrated in the high D-waves (worst pull: 1^3D_3, the $\psi_3(3842)$, -51 MeV).
- Not overfitting — the cross-validation (next) confirms the parameters generalize.

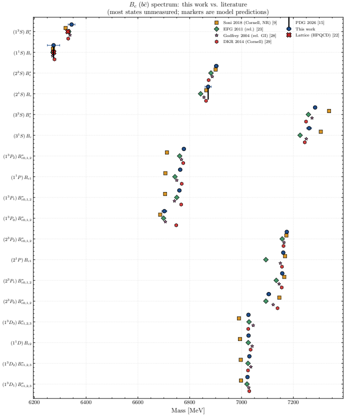
Charmonium vs. the Literature



Per-state deviation $M_{\mathrm{model}} - M_{\mathrm{exp}}$: this work vs. six literature models; green band $= \pm 10$ MeV.

Results: B_c — Pure Predictions

- B_c**:** the two known pseudoscalars are reproduced exactly (the sector is exactly determined); the model then predicts the **entire B_c tower** — S, P and D-waves.
- The mixed-flavour B_c sector is a real test: *genuine predictions*, consistent with dedicated relativistic models.

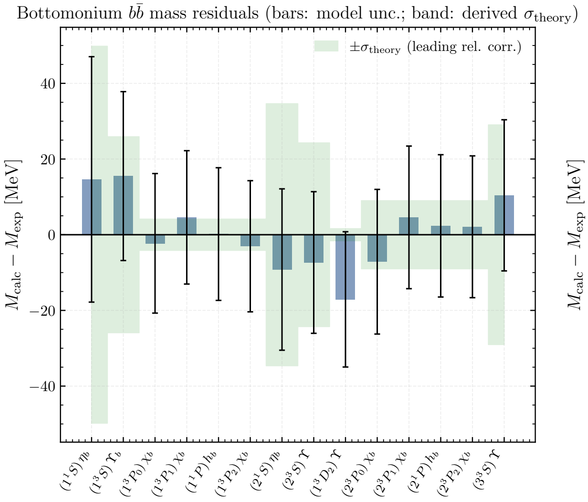


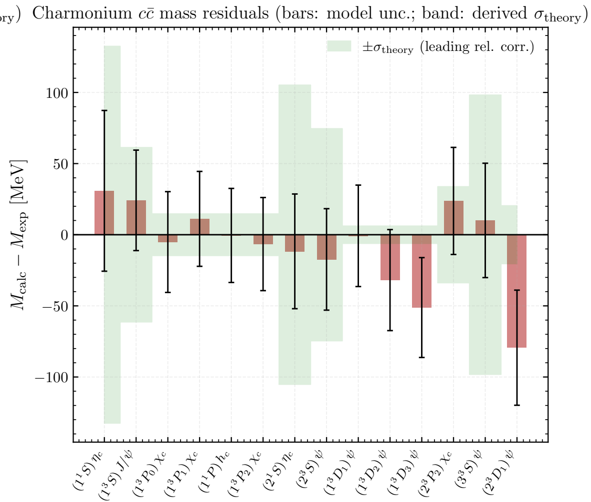
B_c levels: this work vs. literature predictions.

Fit Quality: Residuals & Pulls

- The handful of charmonium D-wave outliers carry the large pulls — the same relativistic tension seen in the χ^2.
- Most pulls lie within $\pm 2\sigma$: the error model is honest, not inflated to flatter the fit.

Mass Residuals per State





$M_{\text{calc}} - M_{\text{exp}}$ per state, with model bars and the $\pm 10\,\text{MeV}$ band.

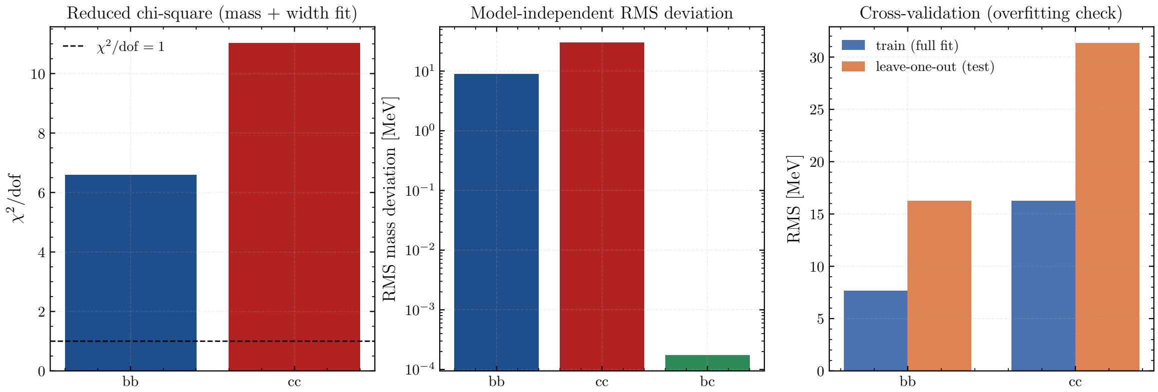
Overfitting Check: Leave-One-Out Cross-Validation

Test: refit the potential on every level *except* one, predict the held-out level, and repeat.

- $b\bar{b}$: LOO 11.1 vs full-fit 5.6 MeV.
- $c\bar{c}$: LOO 31.0 vs full-fit 14.1 MeV.

A ratio $\sim$2 with so few levels per free parameter is normal; held-out levels are predicted about as well as fitted ones $\Rightarrow$ the rigid **4-parameter** Cornell form **generalizes** — it is not memorizing the data.

Cross-Validation & Goodness of Fit



χ^2/dof, RMS deviation, and train/test RMS per sector.

6. Results: Decays & Static Properties

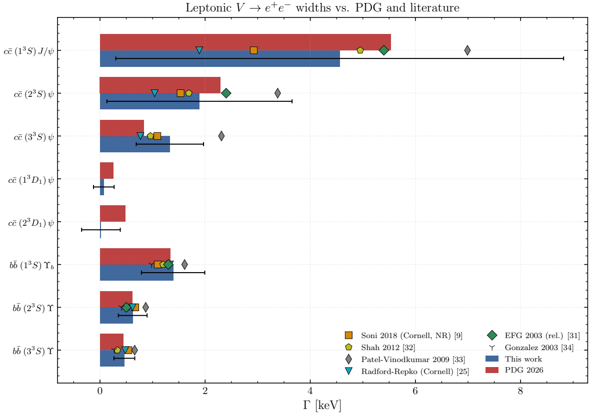
Decay Widths from the Same Wavefunctions

No new parameters. The decay observables are built from the *same* fitted masses and GEM wavefunctions that produced the spectrum — the spectroscopy and the decays are one model, not two.

- **Leptonic** $V \to e^+e^-$ — Van Royen–Weisskopf, $\propto |R(0)|^2$.
- **Two-photon** $P \to \gamma\gamma$ — also $\propto |R(0)|^2$.
- **Radiative** $E1$ and $M1$ transitions between levels.

Leptonic Widths: $V \to e^+e^-$

- Ground and excited Υ and ψ widths agree with PDG at the **10–25%** level.
- $\Upsilon(1S)$: 1.39 vs 1.34 keV; J/ψ: 4.56 vs 5.53 keV.
- The agreement validates $|R(0)|^2$ — the same quantity that feeds every annihilation width.

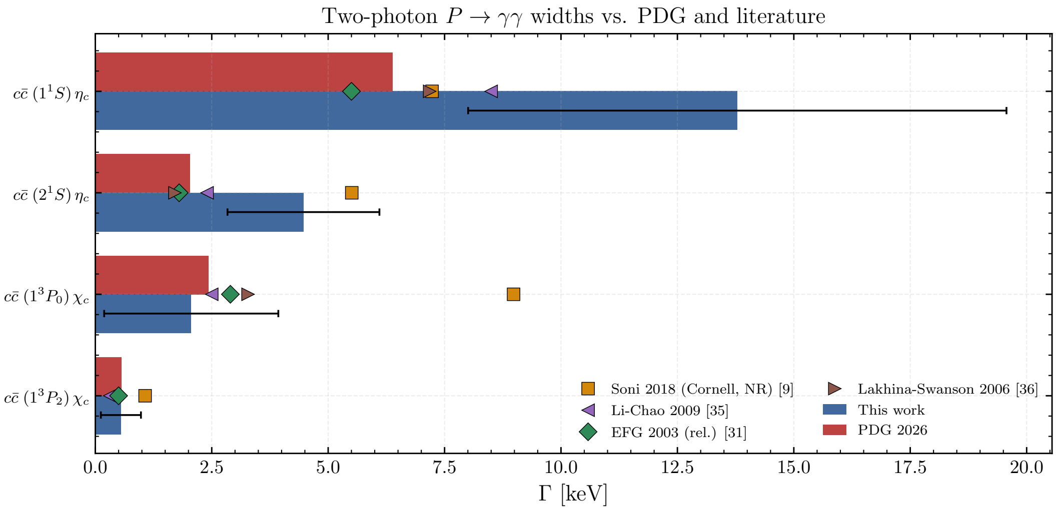


Leptonic $V \to e^+e^-$ widths, this work vs. PDG.

Two-Photon Widths & an Honest Outlier

- The χ_c and χ_b $\gamma\gamma$ widths come out reasonable; but $\eta_c \to \gamma\gamma$ **overshoots by** $\times 2.2$ (13.8 vs 6.4 keV).
- **Diagnosis:** the non-relativistic *over-concentration* of $|R(0)|^2$ for the light, relativistic charm S-wave — a known NR-limit artefact, *not* a GEM/basis error.
- It is the *same* physics that inflates the charmonium mass χ^2: the model quantitatively flags its own limit.

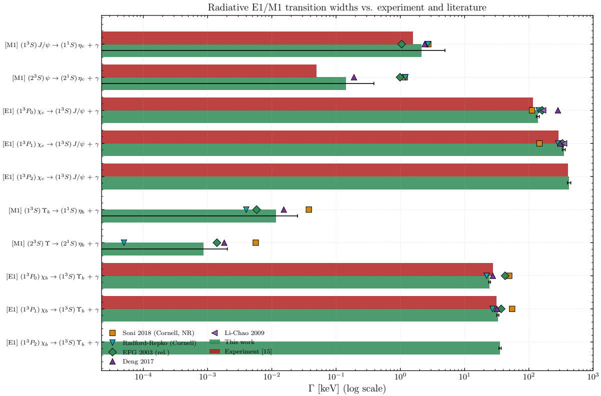
Two-Photon Widths



Two-photon $P \to \gamma\gamma$ widths vs. PDG.

Radiative Transitions: $E1$ & $M1$

- $E1$ ($\chi_c \to J/\psi\,\gamma$, $\chi_b \to \Upsilon\,\gamma$) reproduce experiment well — e.g. $\chi_{b2} \to \Upsilon$ at 37.5 keV.
- $M1$ ($J/\psi \to \eta_c\,\gamma$) are tiny and notoriously model-sensitive: order-of-magnitude agreement with a large, honest uncertainty band.
- Together they probe the wavefunction *overlaps* between levels — a stiffer test than the spectrum alone.

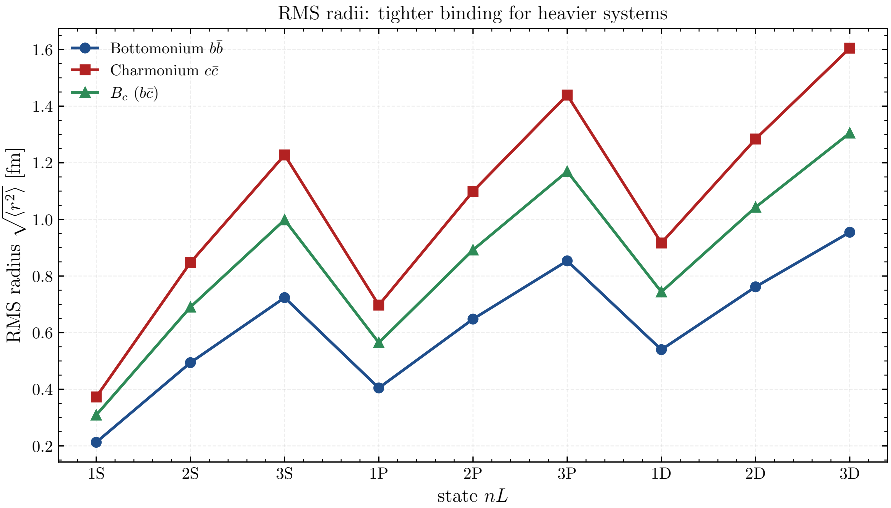


$E1/M1$ transition widths vs. experiment (log scale).

Static Size: RMS Radii

- $\sqrt{\langle r^2 \rangle}$ grows with radial excitation n and is *smaller* for the heavier systems ($b\bar{b}\,1S \approx 1.1$ vs $c\bar{c}\,1S \approx 1.9\ \mathrm{GeV}^{-1}$).
- Tighter bottomonium binding is exactly what underlies its larger $|R(0)|^2$ and leptonic widths — the picture is internally consistent.
- A free by-product of the same eigenvectors: no extra computation.

RMS Radii per State



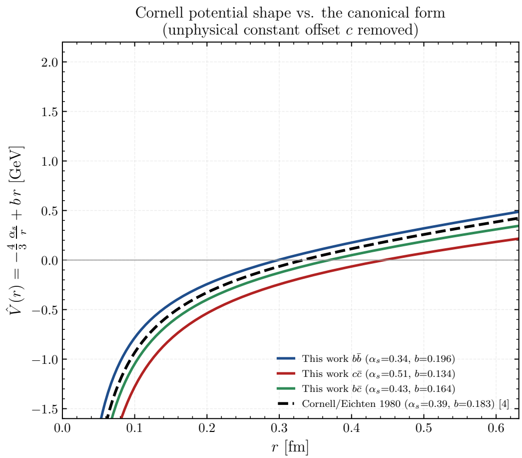
RMS radii $\sqrt{\langle r^2 \rangle}$ per state, per sector.

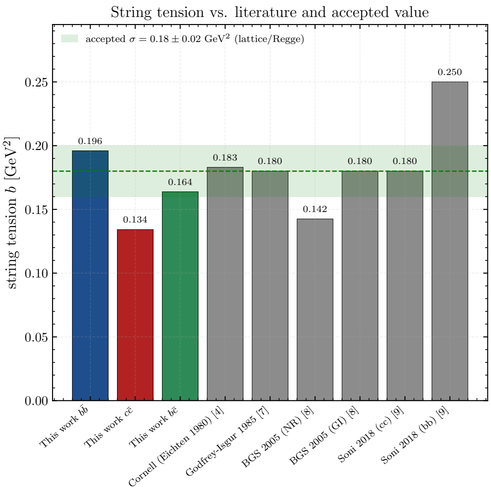
7. Validation & Outlook

Validation Against the Literature

- **The potential is physical:** the fitted string tension $b \approx 0.13$–$0.20\ \mathrm{GeV}^2$ brackets the accepted lattice/Regge value $\sigma \approx 0.18\ \mathrm{GeV}^2$, and the Cornell shape overlays the canonical Eichten 1980 potential.
- **Accuracy among published models:** the aggregate RMS sits with dedicated relativistic models — bearing in mind this work is *fitted* to these masses, so the meaningful comparison is the spread of the genuine-prediction models.

Cornell Shape & String Tension vs. the Literature





Cornell shape vs. Eichten 1980; string tension vs. the accepted $\sigma \approx 0.18 \pm 0.02\ \mathrm{GeV}^2$.

Conclusions

- **One analytic framework:** a closed-form Cornell + GEM solver reproduces bottomonium to **7 MeV** and charmonium to **29 MeV**, cross-validated as *not* overfitting.
- **Spectrum and decays from the same wavefunctions:** leptonic, two-photon, radiative and open-flavour widths — and the lone clear outlier ($\eta_c\,\gamma\gamma$) *quantitatively* marks the non-relativistic limit rather than a numerical fault.
- **Predictive reach:** the rigid four-parameter form extends to the mixed-flavour B_c and D, predicting **dozens** of yet-unobserved states.

Future Outlook

- **Complete the catalogue:** extend the decay-width and RMS-radius tables uniformly across all three sectors and all predicted levels.
- **Refine the charm sector:** add relativistic / coupled-channel corrections to tame the $c\bar{c}$ tension and the $\eta_c \to \gamma\gamma$ overshoot.
- **Three-body extension:** promote the same closed-form GEM machinery to the **three-body problem** in Jacobi coordinates — opening the door to heavy *baryons*.

One analytic, validated potential model spans $b\bar{b} \to c\bar{c} \to B_c \to D$.